

Transaction Sets from Transaction Pattern

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Abstract

While analyzing business situations at analysis level, it is a challenging activity to identify appropriate objects and their relationships. For such purpose different patterns and data models are used, but most of them deal in few specific situations. Thesis of this research is to identify such pattern that covers most of the business transactions. Hence transaction pattern is identified. Research further identifies different transaction types by grouping different objects of the pattern, where each group is representing different transaction type that fulfills

Key words

Transaction types, Transaction Pattern, Players

1. Introduction

A transaction is a recording or logging of any event of significance; hence, transaction is “a significant event remembered” [1]. In business scenarios, event that involve currency, count, or date time are classified as a transaction.

Examples:

Deposit, order, payment, purchase, refund, rental, reservation, sale and withdrawal are examples of currency based transactions.

Courses enrollment, attendance register, books issued or returned and marks obtained are examples of count based transactions.

Time in time out, birth certificate, appointment, schedule, session logs are examples of date time based transactions.

While defining transactions multiple objects are used. These objects vary from transaction to transaction. To meet this objectives, a generalized business model is required that captures a wide range of business transactions and their objects.

Existing business models such as **Transactions and Accounts**[2] [3] [4], **Banking Patterns** [3] and **Business Models** of data model resource book [6] are domain specific, and lacked sufficient generality. However, Peter Coad's *transaction pattern* [1] is a higher-level description of a generic

business process (or a template) that has helps in the analysis and design of a wide range of business domains.

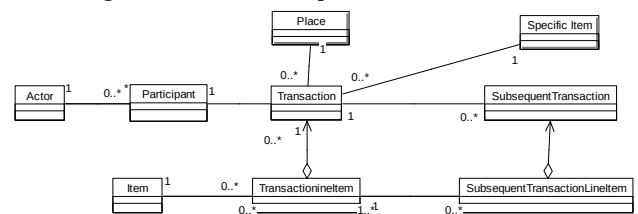
Research selects transaction pattern, which supports large variety of business transaction situations. By arranging different transaction objects, different type of transactions are identified and named them *transaction sets*. Research is helpful for analyzing business applications and specially in designing different business processes. Different transaction sets are useful in validating transaction objects of specific business situation and validation their relationships.

2. Transaction Sets

Transaction pattern is an analysis pattern that specifies a transaction with the help of players. *Players* are objects with special roles and associations in a transaction. Players include actor, participant, place, transaction line item, item, specific item, subsequent transaction, and subsequent transaction line item [1]. A workflow consists of a sequence of transactions in which a transaction is followed by a subsequent transaction. In *figure-1* transaction pattern, with its players and relationship is presented.

Analysis of business applications indicates that not all transaction contain all players. It is also observed simplest transaction is based on only two players and that are participant and transaction. The set of players varies from transaction to transaction.

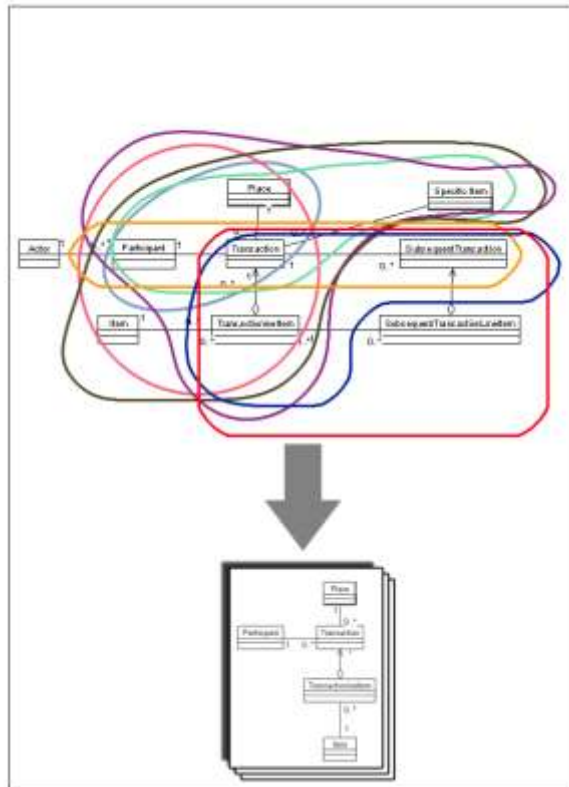
Figure-1 Transaction pattern



The research defines *transaction set* by grouping of players relating to different type of transactions. In some transaction few players are involved such as an appointment transaction requires three players (participant, transaction and place), where

participant is patient, transaction is appointment and place is clinic. There are some transaction that involve more players such as order transaction which requires five players (participant, transaction, transaction line item, item, and place), where participant is customer, transaction is order, transaction line item is order detail and place is factory. By varying the selection of players, different transaction sets are generated.

Figure 2 Transaction sets from transaction pattern



The phenomenon of grouping different players and defining their transaction set is shown in figure-2. Different transaction sets represent different analysis and design situation. By varying the selection of players different transaction types are generated. Transaction pattern is a superset that contains many subsets of transaction sets. Each subset is representing a separate transaction type.

Eight transaction sets are presented and are divided in three broad categories. There can be more transaction sets by varying players.

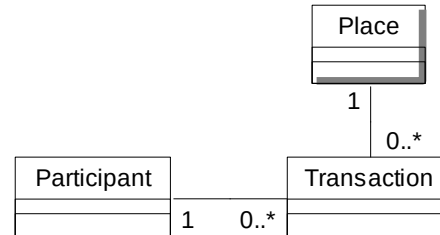
Transaction based on participant

Transaction based on items

Transaction based on subsequent transaction.

2.1 Transaction based on participant

A simple transaction may contain a few players such as participant and transaction, where place is an optional player.



Examples:

Patient - appointment - clinic

Student - attendance

Visitor - gate pass – department

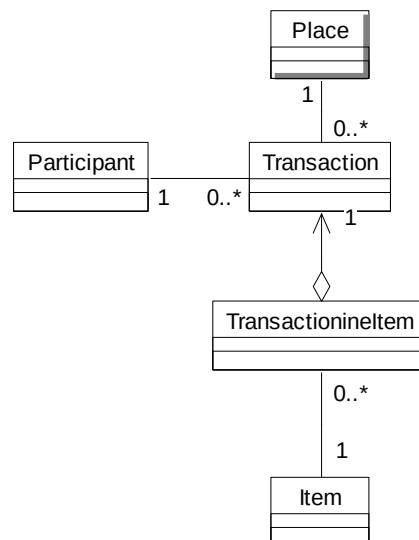
Driver – driving license

2.2 Transaction Based on Items

Second type of transaction is based on items. Item is further classified as general item or specific item.

General item transaction

These transactions are based on item and typically involve a count or quantity of the item. Players of these transactions are participants, transaction, transaction line item, item and place.



Examples:

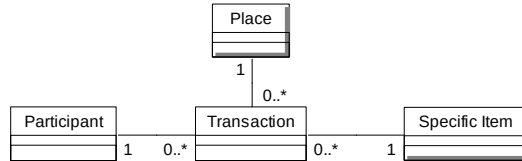
Customer- order –order line item -product

Buyer –payment –payment line item -goods

Vendor – purchase- purchase line item - products

Specific item transaction

These transactions are based on items whose individual identity is important to track such as the serial number. These transactions are designed for a single item and do not involve quantity as in general item transaction. Players of these transactions are participant, transaction and specific item, where place is an optional player.

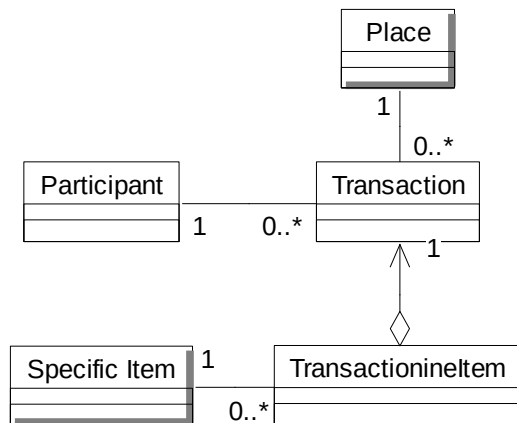


Examples:

Investor- payment- Specific aircraft
 Agent – loading - specific container - port
 Shipper - shipment - specific ship,
 Buyer – rental- specific vehicle
 Member – subscribe –specific credit card

Specific item transaction with line item

These transactions support multiple specific items. These transactions use transaction line item to store list of specific items. Players of these transactions are participant, transaction, transaction line item and specific item where place is optional.

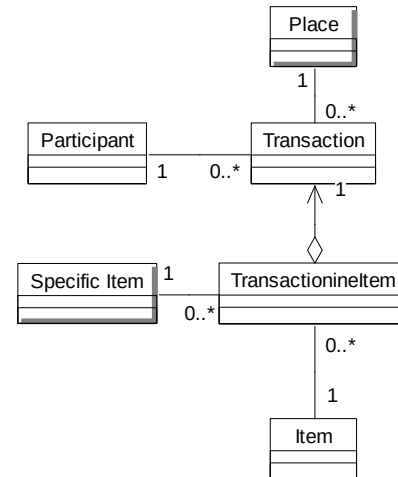


Examples:

Student – issue –issue line item –specific book
 Customer –purchase –purchase line item – electric goods
 Account holder – deposit –deposit line item – cash bonds - bank branch
 Buyer – purchase –purchase line item – computer accessories

General and specific item transaction

These transactions are based on both kind of items, general items and specific items. These transactions use line item to store list of specific items and general item. Players of these transactions are participant, transaction, transaction line item, specific item, item and place.



Examples:

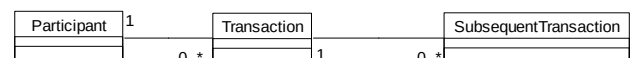
Customer –purchase –purchase line item – electric goods - goods
 Buyer – purchase –purchase line item – computer accessories – printer pages -department
 Customer- order – order line item – product – specific product

2.3 Transaction based on subsequent transaction.

These transactions are based on two transactions, one existing and other subsequent transaction. There are three possible combinations.

Transaction and subsequent transaction both without line item

These types of transactions involve simple transaction and subsequent transaction. Players are participant, transaction and subsequent transaction. The relationship between these players is many to many. In relational data model there must exist association table.



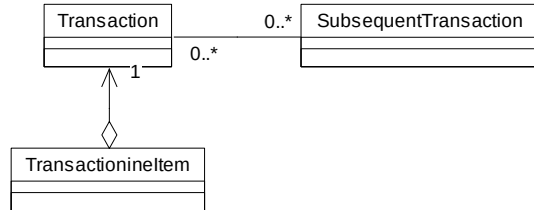
Examples:

Patient – appointment – treatment /admission
 Student - attendance – examination
 Visitor - gate pass – entry

Driver – driving license- renewal

Transaction with line item and subsequent transaction without line item

These types of transactions involve transaction with line item and subsequent transaction without line item. Players are transaction, transaction line item and subsequent transaction. The relationship between transaction and subsequent transaction player is many to many.



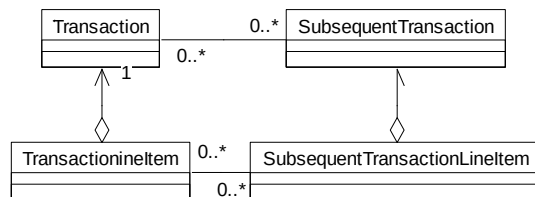
Examples:

Order –order line item -payment

Purchase- purchase line item - packing

Transaction and subsequent transaction both with line item

These types of transactions involve transaction and subsequent transaction both with line item. Players are transaction, transaction line item, subsequent transaction and subsequent transaction line item. The relationship among transaction, subsequent transaction and their line items is many to many. In relational data model there must exist association tables.



Examples:

Order –order line item –payment – payment line item

Payment –payment line item –shipment-shipment line item

Purchase- purchase line item – packing –packing line item

3. Attributes

Each Player contain attributes Essential attributes of each player are given in *table-1*

Table-1 Attributes of players

Players	Attributes
Actor	Name, address, phone
Participant	Number, start date, end date,

	authorization level, password
Transaction	Number, date, time, status
Transaction line item	Number, quantity, status
Item	Number, name , description, price, default value
Specific item	Serial number, purchase date, custom value
Place	Number, name, address

Transaction pattern defines only essential attributes that significantly defines the role of player. An attribute **AboutMe** is available in every player that indicates that other user defined fields can be added to further describe the player.

4. Behaviors

Each Player contain behaviors. Transaction pattern defines set of behaviors specialized for data retrieval or to perform calculation. They are classified in three broad categories 1-value check, 2-single record calculation, 3-calculation over collection.

For **value check** transaction pattern provides behaviors such as isActive, isAuthorized, isStatusValue, CheckStatus, getCustomElseDefault. In general there are two main behaviors is<value> and check<value> where as value can be Boolean, other datatype or some user defined value.

In **single record calculation** there are two behaviors CalcForMe, RateMe, these are defined in each player. These are used to perform calculation on single record or among fields of tables. CalcForMe is used to perform different arithmetic, datatype specific or business operation. RateMe is used to sort data ascending or descending order based on user defined criteria or SQL (WHERE clause).

In **calculation over collection** there are four behaviors howMuch, howMany, Rank<AssociatedPlayer>, calOver<AssociatedPlayer>. These behaviors are performed on set of records of single table or multiple tables. These operations are usually based on user defined condition or in SQL (GROUP BY, HAVING, and use of aggregate operations (SUM, COUNT, MIN, MAX, etc.)). howMuch and howMany are defined for each player and are used for single table. Rank<AssociatedPlayer> and calOver<AssociatedPlayer> are also defined in each player and are used for associated player such as transaction is associated with participant therefore calOverTransaction is defined in participant. Where as *AssociatedPlayer* can be actor, participant, transaction,

transaction line item, subsequent transaction, subsequent transaction line item, item, specific item, place, associate, and other associate

5. Conclusions

With the help of identified transaction sets, searching and identifying different business transactions becomes easy. The relationship, attributes and behaviors can be used directly as defined in transaction pattern. The research is also helpful in validating and completing the missing objects and their possible attributes.

6. References

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